

Problem

Repeat Example 14.18 for band-pass filter BP_6 .

Example 14.18:

Using the standard 600- Ω resistor used in telephone circuits and a series RLC circuit, design the band-pass filter BP_2 in Fig. 14.65.

Solution:

The band-pass filter is the series RLC circuit in Fig. 14.35. Inasmuch as BP_2 passes frequencies 697 to 852 Hz and is centered at $f_0 = 770$ Hz, its bandwidth is

$$B = 2\pi(f_2 - f_1) = 2\pi(852 - 697) = 973.89 \text{ rad/s}$$

From Eq. (14.39),

$$L = \frac{R}{B} = \frac{600}{973.89} = 0.616 \text{ H}$$

From Eq. (14.27) or (14.55)

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{4\pi^2 f_0^2 L} = \frac{1}{4\pi^2 \times 770^2 \times 0.616} = 69.36 \text{ nF}$$

Figure 14.65: Block diagram of detection scheme.

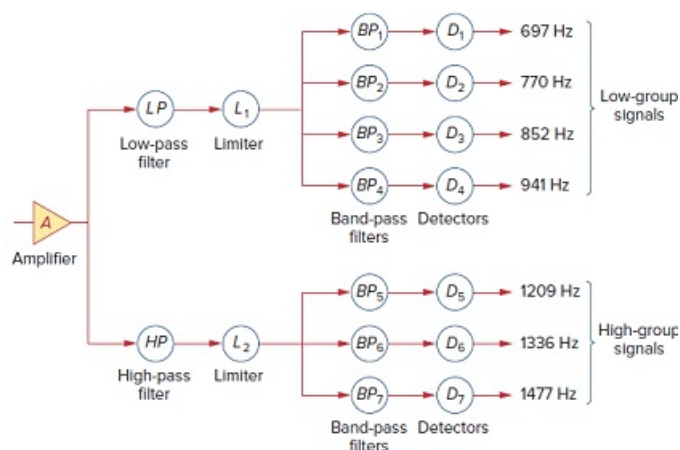
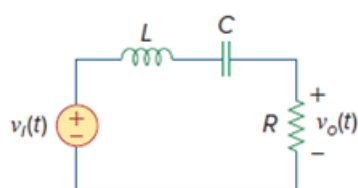


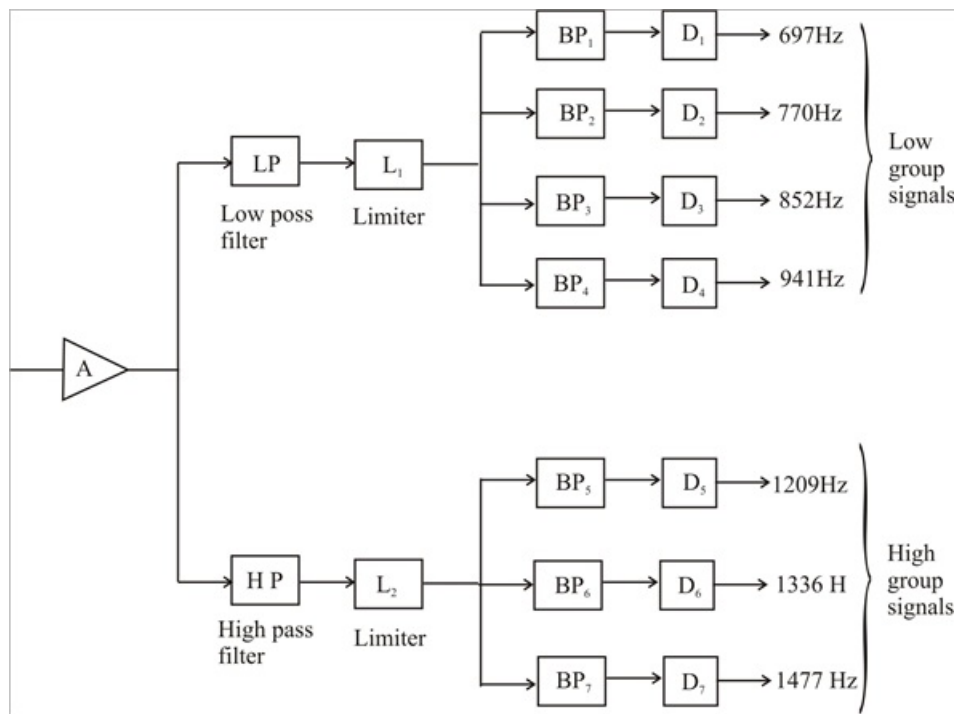
Figure 14.35: A band-pass filter.



Step-by-step solution

Step 1 of 3

Given figure,



Step 2 of 3

Given that $R = 600\Omega$

The band pass filter is the series RLC circuit

Since BP_6 passes frequencies 1209 Hz to 1477 Hz and is centered at $f_0 = 1336$ Hz. Its band width is,

$$B = 2\pi(f_2 - f_1)$$

$$B = 2\pi(1477 - 1209)$$

$$\therefore B = 1.6839 \text{ k rad / sec}$$

$$\text{Band width } (B) = \frac{R}{L}$$

$$L = \frac{R}{B}$$

$$L = \frac{600}{1.6839 \times 10^3}$$

$$\therefore \boxed{L = 0.356 \text{ H}}$$

Step 3 of 3

Resonant frequency $(\omega_0) = \frac{1}{\sqrt{LC}}$

$$2\pi f_0 = \frac{1}{\sqrt{LC}}$$

$$(2\pi f_0)^2 = \frac{1}{LC}$$

$$C = \frac{1}{(2\pi f_0)^2 \times L}$$

$$C = \frac{1}{(2\pi \times (1336))^2 \times 0.356}$$

$$\therefore \boxed{C = 39.86 \text{ nF}}$$